GRADE 10

Consolidation worksheet

1. A(-4; -1), B(3; 2) and C(2; 3) are the vertices of ΔABC. D is a point of AB so that CD $\perp$ AB. E is a point on the y-axis and H and K are on the x-axis. Determine:

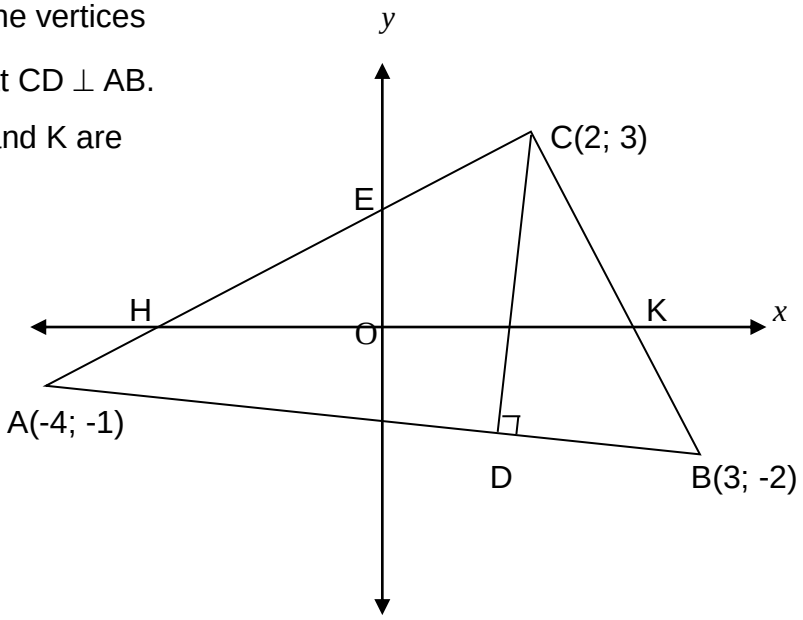
   a) the equation of AC

   b) the equation of BC

   c) the equation of CD

   d) the length of OE

   e) the length of HK



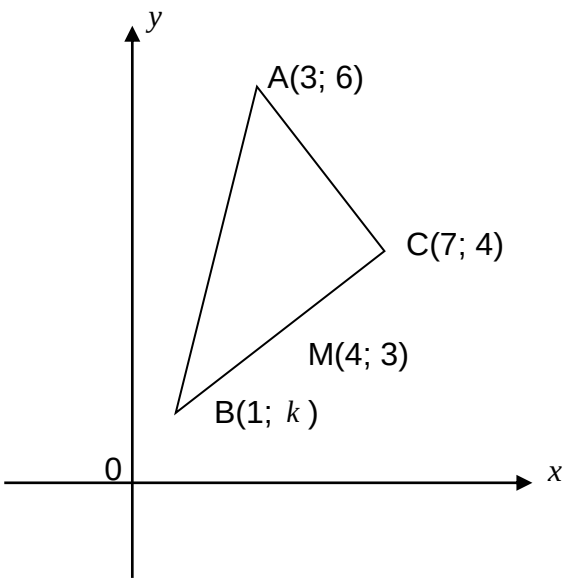
2. In the figure, A(3; 6) , B(1; k) and C(7; 4) are the vertices of a triangle with M(4; 3) the midpoint of BC.

   a) Show that $k = 2$.

   b) If R is the midpoint of AC, show using analytical methods that:

      i) MR = ½ BA

      ii) MR $\parallel$ BA



3. In the figure, A(2; 1), B(8; 5) and C(4; 8) are the vertices of a triangle. The point M(x; y) lies on the y-axis such that CM is parallel to BA.

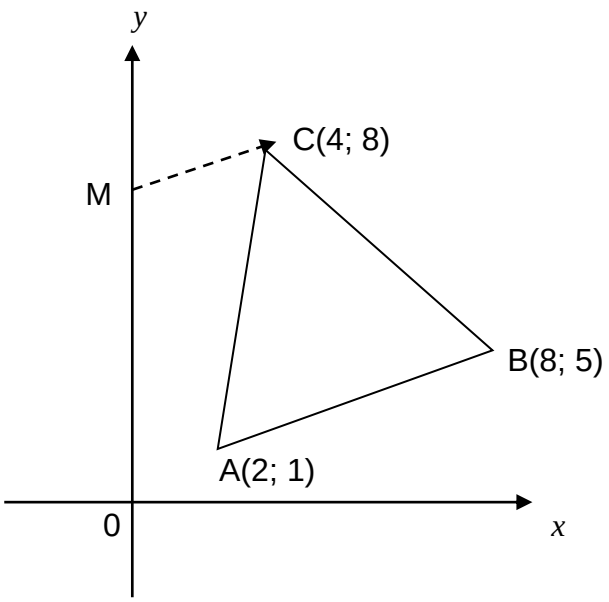
   Determine the following:

   a) the length of BC

   b) the gradient of AB

   c) the equation of CM, in the form y=mx + c

   d) the co-ordinates of M



4. The equation of a given straight line is $y = px + 6$.

Determine the value of p in each of the following cases:

a) the given line [$y = px + 6$] passes through the point (-1; -4).

b) the given line is // to the line $2y + x = 4$.

c) the given line is // to the x-axis.

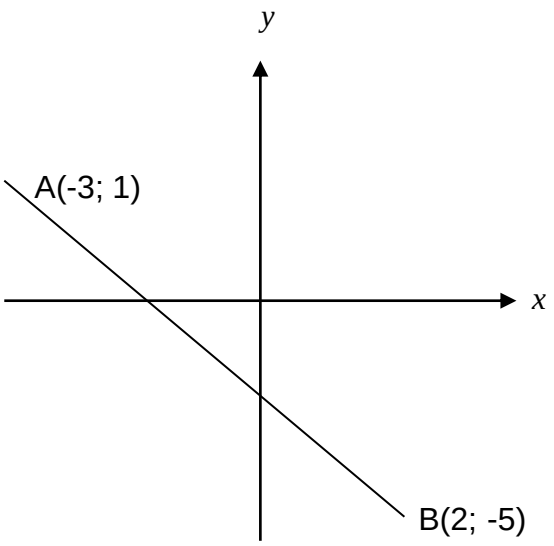
5. Find the equation of the line passing through (-1; 1) and perpendicular to the line $x - 3y = 5$.

6. M(8 ½ ; -1) is the midpoint of line segment xy with $x(4a; b)$ and $y(5; -3)$.

Determine the values of a and b.

7. In the sketch the points A(-3; 1) and B(2; -5) are indicated in a Cartesian plane.

The points C(m; n) and D(1; p) are given, but are not indicated.



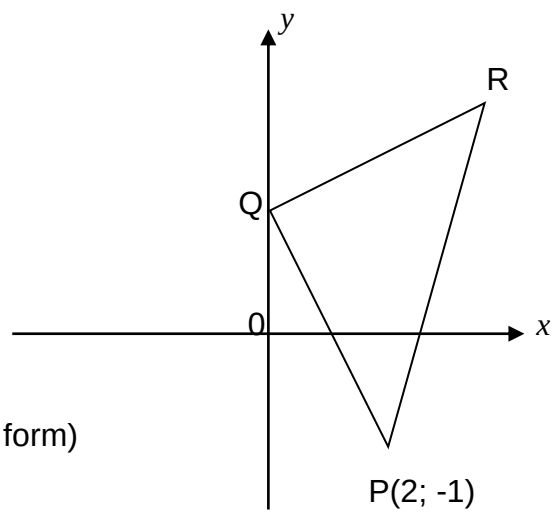
a) If C is the midpoint of AB, calculate the value of m and n.

b) If A, B and D are collinear, calculate the value of p.

c) Determine the equation of the straight line which is parallel to AB and which passes through the origin.

8. In the figure, the co-ordinates of P are (2; -1).

The equation of QR is $x - 2y + 6 = 0$ and the equation of PR is given by $3x - y - 7 = 0$.



a) What is the gradient of QR?

b) Find the co-ordinates of Q.

c) Prove that $P\hat{Q}R = 90°$.

d) Determine the co-ordinates of R.

e) Find the length of PR. (leave answer in surd form)